

# Final Exam Review

*Based on Alpha C. Chiang and Kevin Wainwright (4th Edition)*

Material: Functions, Linear Models, Differentiation, Optimization, Exponential & Logarithmic Functions, Constrained Optimization, Envelope Theorem, Integration, Dynamics

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## Functions and Graphs

### Definitions

- A function assigns to each input  $x$  exactly one output  $y = f(x)$ .
- Domain: all permissible  $x$ -values. Range: the set of resulting  $y$ -values.

### Common Function Types

Linear:  $y = a + bx$ ,      Quadratic:  $y = ax^2 + bx + c$ ,      Power:  $y = ax^b$ .

Exponential:  $y = Ae^{bx}$ ,      Logarithmic:  $y = a + b \ln x$ .

### Economic Uses

- Derivatives represent marginal changes (marginal cost, marginal utility).
  - Elasticity measures percentage responsiveness.
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## Linear Functions

### General Form:

$$y = a + bx.$$

Slope:  $b = \frac{\Delta y}{\Delta x}$ .

### Applications

- Demand:  $Q_d = a - bP$
- Supply:  $Q_s = c + dP$

- Market equilibrium:  $Q_d = Q_s$ .

### Elasticity

$$E = \frac{dy}{dx} \cdot \frac{x}{y}.$$

Note: Elasticity varies along a linear function.

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## Differentiation

### Key Rules

$$\frac{d}{dx}x^n = nx^{n-1}, \quad (uv)' = u'v + uv', \quad \left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}.$$

$$\text{Chain Rule: } (f(g(x)))' = f'(g(x)) \cdot g'(x).$$

### Economic Interpretation

- $MC = C'(Q)$ ,  $MR = R'(Q)$ .
  - Profit maximization requires  $MR = MC$ .
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## Optimization of Single-Variable Functions

To find extrema:

$$f'(x) = 0 \quad (\text{FOC}).$$

### Second-Order Test

- If  $f''(x) < 0$ : maximum.
- If  $f''(x) > 0$ : minimum.

**Inflection Point:** when  $f''(x) = 0$  and concavity changes sign.

### Examples

- Profit maximization  $\pi(Q) = R(Q) - C(Q)$ .
  - Cost minimization with single variable.
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# Exponential and Logarithmic Functions

## Exponential Form

$$y = Ae^{bx} \Rightarrow y' = by.$$

Proportional growth rate is constant:

$$\frac{y'}{y} = b.$$

## Logarithmic Function

$$y = a + b \ln x \Rightarrow y' = \frac{b}{x}.$$

## Elasticity

$$E = \frac{dy}{dx} \cdot \frac{x}{y}.$$

Log-linear forms have constant elasticity.

## Economic Applications

- Continuous compounding:  $A = Pe^{rt}$ .
  - Growth models:  $Y(t) = Y_0 e^{rt}$ .
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# Optimization of Multi-Variable Functions

## First-Order Conditions

$$f_x(x, y) = 0, \quad f_y(x, y) = 0.$$

## Hessian Matrix

$$H = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}.$$

## Classification

- If  $\det H > 0$  and  $f_{xx} < 0$ : maximum.
- If  $\det H > 0$  and  $f_{xx} > 0$ : minimum.
- If  $\det H < 0$ : saddle point.

## Economic Interpretations

- Diminishing marginal returns.
  - Substitutability from cross-partials.
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## Optimization with Equality Constraints

Maximize  $f(x, y)$  subject to  $g(x, y) = b$ .

### Lagrangian

$$\mathcal{L}(x, y, \lambda) = f(x, y) - \lambda(g(x, y) - b).$$

### FOCs

$$f_x = \lambda g_x, \quad f_y = \lambda g_y, \quad g(x, y) = b.$$

**Interpretation**  $\lambda$  is the shadow price: the marginal value of relaxing the constraint.

### Bordered Hessian

$$H_B = \begin{bmatrix} 0 & g_x & g_y \\ g_x & f_{xx} & f_{xy} \\ g_y & f_{xy} & f_{yy} \end{bmatrix}.$$

Sign of  $\det(H_B)$  determines max/min.

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## Envelope Theorem and Duality

**Envelope Theorem** If  $v(b)$  is the maximum of  $f(x)$  subject to  $g(x) = b$ , then

$$\frac{dv}{db} = \lambda^*.$$

This simplifies comparative statics by avoiding differentiation of optimal choices.

### Duality

- Utility maximization vs cost minimization.
  - Both yield the same marginal rate of substitution conditions.
  - Shadow prices appear in both problems.
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# Integration

## Basic Rules

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad \int e^{ax} dx = \frac{1}{a}e^{ax} + C, \quad \int \frac{1}{x} dx = \ln |x| + C.$$

## Definite Integral

$$\int_a^b f(x) dx = F(b) - F(a).$$

## Economic Uses

- Consumer surplus.
  - Present value:  $\int_0^\infty e^{-rt} R(t) dt$ .
  - Capital accumulation solved by integration.
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# Economic Dynamics

## Differential Equations

$$\dot{y} + ay = b \quad \Rightarrow \quad y(t) = Ce^{-at} + \frac{b}{a}.$$

## Growth Models

$$Y(t) = Y_0 e^{rt}.$$

## Decay Models

$$K(t) = K_0 e^{-\delta t}.$$

Dynamics describe continuous-time adjustment processes in economics (capital, population, debt).

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## Summary Table

| Topic                     | Mathematical Tool      | Economic Interpretation         |
|---------------------------|------------------------|---------------------------------|
| Slopes                    | Derivative             | Marginal change                 |
| Concavity                 | Second derivative      | Diminishing returns             |
| Elasticity                | Log derivative         | Responsiveness                  |
| Multivariate optimization | Hessian matrix         | Classification of behavior      |
| Constraints               | Lagrange multipliers   | Shadow values                   |
| Comparative statics       | Envelope theorem       | Value-function sensitivity      |
| Integration               | Antiderivatives        | Accumulation, surplus           |
| Dynamics                  | Differential equations | Growth, decay, adjustment paths |